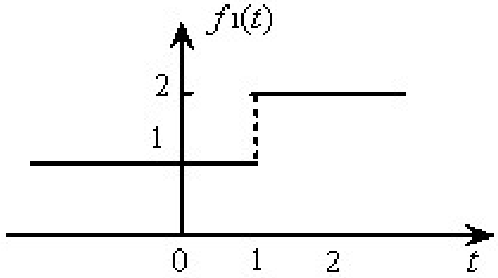
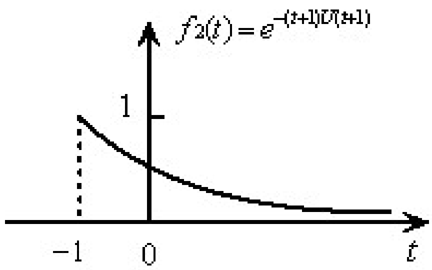
检测技术基础

第 5、6、7 讲作业参考解答

第 5 讲作业

5－1 求 **y(t)=f$_1$(t)*f$_2$(t)**，并画出 **y(t)**的波形

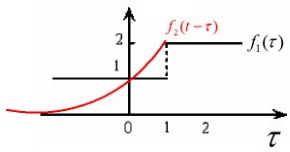




解：

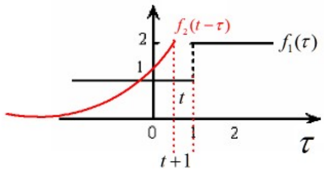
$$f_1(t)=1+U(t+1)$$
$$f_2(t)=e^{-(t+1)}U(t+1)$$

变量替换，反折



$t=0$

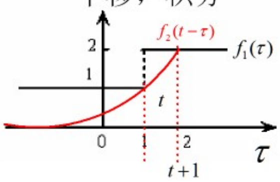
平移，积分



$t\le 0$

$$f_1(t)*f_2(t)=\int_{-\infty}^{t+1}e^{-(t-\tau+1)}\mathrm{d}\tau=\int_{-\infty}^{t+1}e^{\tau-t-1}\mathrm{d}\tau\overset{u=\tau-t-1}{=}\int_{-\infty}^{0}e^{u}\mathrm{d}u=1$$

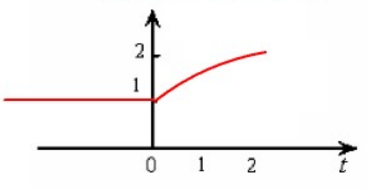
平移，积分



$t>0$

$$f_1(t)*f_2(t)=\int_{-\infty}^{1}1\cdot e^{-(t-\tau+1)}\mathrm{d}\tau+\int_{1}^{t+1}2\cdot e^{-(t-\tau+1)}\mathrm{d}\tau$$
$$=\int_{-\infty}^{t+1}1\cdot e^{-(t-\tau+1)}\mathrm{d}\tau+\int_{1}^{t+1}1\cdot e^{-(t-\tau+1)}\mathrm{d}\tau$$
$$\overset{u=\tau-t-1}{=}\int_{-\infty}^{0}1\cdot e^{u}\mathrm{d}u+\int_{-t}^{0}1\cdot e^{u}\mathrm{d}u$$
$$=1+1-e^{-t}=2-e^{-t}$$

$f_1(t)*f_2(t)$的图形



5-2 证明自相关函数的性质：

1） $R_x(0)$为自相关函数的最大值：$R_x(0)\ge R_x(\tau)$

2） 自相关函数的取值范围：$\mu_x^2-\sigma_x^2\le R_x(\tau)\le\mu_x^2+\sigma_x^2$

3） $R_x(-\tau)=R_x(\tau)$ $R_x(\tau)$是τ的偶函数

证明：

1) $R_x(0)$为$R_x(\tau)$的上界：$R_x(0)\ge R_x(\tau)$ （许瓦尔兹不等式：$E^2(XY)\ge E^2(X)E^2(Y)$）

证明:$R_x{}^2(\tau)=E^2\left(X(t).X(t-\tau)\right)\le E\left(X^2(t)\right)E\left(X^2(t-\tau)\right)=E\left(X^2(t)\right)E\left(X^2(t)\right)$
$=R_x(0)R_x(0)=R_x^2(0)$

2) 自相关函数的取值范围：$\mu_x^2-\sigma_x^2\le R_x(\tau)\le\mu_x^2+\sigma_x^2$

$$\rho_x(\tau)=\frac{E\left[\left(x(t)-\mu_x\right)\left(x(t-\tau)-\mu_x\right)\right]}{\sigma_x\sigma_x}=\frac{R_x(\tau)-\mu_x^2}{\sigma_x^2}$$

$\therefore\left|\rho_x(\tau)\right|\le 1$

$\therefore\left|\frac{R_x(\tau)-\mu_x^2}{\sigma_x^2}\right|\le 1$

$\therefore\mu_x^2-\sigma_x^2\le R_x(\tau)\le\mu_x^2+\sigma_x^2$

3) $R_x(-\tau)=R_x(\tau)$ $R_x(\tau)$是τ的偶函数

证明：

$$R_x(\tau)=\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}x(t)x(t-\tau)dt=E\left(X(t).X(t-\tau)\right)$$

$$R_x(-\tau)=E\left(X(t).X(t+\tau)\right)$$

令$t'=t+\tau$

则：

$$R_x(-\tau)=E\left(X(t'-\tau).X(t')\right)=E\left(X(t-\tau).X(t)\right)=E\left(X(t).X(t-\tau)\right)=R_x(\tau)$$

5-3 证明自相关函数的性质：

1）互相关函数的取值范围:$\left(\mu_x\mu_y-\sigma_x\sigma_y\right)\le R_{xy}(\tau)\le\left(\mu_x\mu_y+\sigma_x\sigma_y\right)$

2）$R_{xy}(-\tau)=R_{yx}(\tau)$

证明：

1）互相关函数的取值范围:$\left(\mu_x\mu_y-\sigma_x\sigma_y\right)\le R_{xy}(\tau)\le\left(\mu_x\mu_y+\sigma_x\sigma_y\right)$

证明：

$$\rho_{xy}(\tau)=\frac{E\left[\left(x(t)-\mu_x\right)\left(y(t-\tau)-\mu_y\right)\right]}{\sigma_x\sigma_y}=\frac{\lim\limits_{T\to\infty}\frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}}\left(x(t)-\mu_x\right)\left(y(t-\tau)-\mu_y\right)\mathrm{d}t}{\sigma_x\sigma_y}$$

$$=\frac{R_{xy}(\tau)-\mu_x\mu_y}{\sigma_x\sigma_y}$$

$$\therefore\left|\rho_{xy}(\tau)\right|\le1$$

$$\therefore\left|\frac{R_{xy}(\tau)-\mu_x\mu_y}{\sigma_x\sigma_y}\right|\le1$$

$$\therefore\left(\mu_x\mu_y-\sigma_x\sigma_y\right)\le R_{xy}(\tau)\le\left(\mu_x\mu_y+\sigma_x\sigma_y\right)$$

2）$R_{xy}(-\tau)=R_{yx}(\tau)$

证明：$R_{xy}(-\tau)=E\left(X(t).Y(t+\tau)\right)=E\left(Y(t+\tau).X(t)\right)$

令$t'=t+\tau$

$$=E\left(Y(t').X(t'-\tau)\right)=E\left(Y(t).X(t-\tau)\right)=R_{yx}(\tau)$$